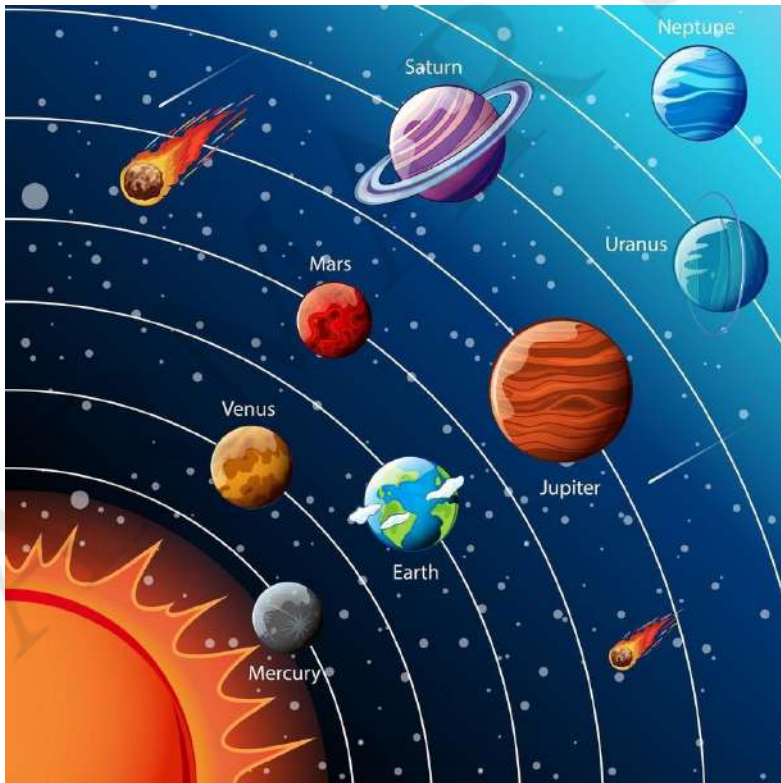


SOLAR SYSTEM





- Nearest galaxy: Andromeda Galaxy
- Study of Universe: Cosmology



ORIGIN OF UNIVERSE

Theories given:

- BIG BANG THEORY



Infinitely hot and dense single point → Exploded

George Lemaitre: 1931

Edwin Hubble

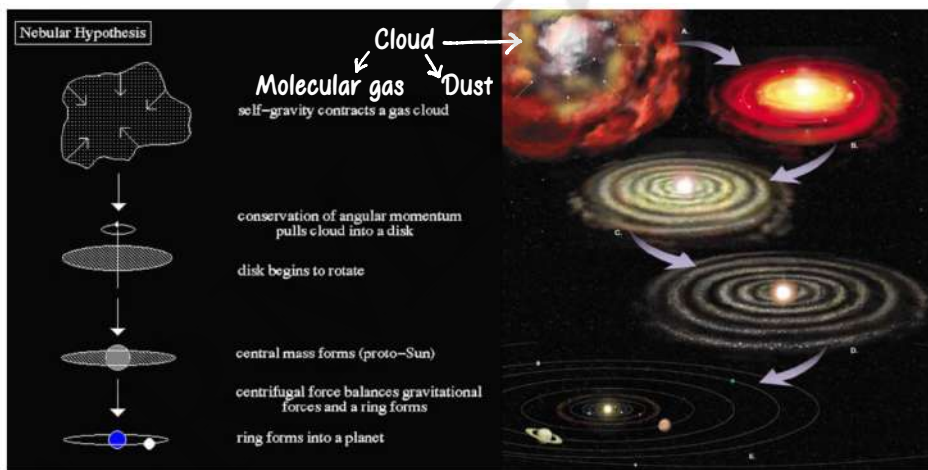
→ Increase in distance of celestial bodies

Origin of Big Bang Theory. Also, → Explosion → 13.6 billion years ago known as Big Bang Explosion

4.5 billion years ago

FORMATION OF OUR SOLAR SYSTEM

- **Nebular Theory, 1755:** by Immanuel Kant
1796: modified by Laplace
- **Nebula:** A giant cloud of dust and gas



- $H_2 + He \rightarrow$ Nuclear Fusion
- $H + H \rightarrow He \rightarrow$ Formation of Sun (mostly made of H_2 and He)
↓
 $H_2 \rightarrow 70\%$

- **Indian Institute of Astrophysics HQ: Bangalore**

CELESTIAL BODIES

Two types:

- **Luminous:** Self-glowing, eg: stars
 - **Non-Luminous:** Not self-glowing, but can reflect light from other sources. Eg: Moon
1. Asteroids: they are small, rocky objects that orbit the Sun
 2. Meteoroids/Meteors: enters Earth's atmosphere and burn up in Mesosphere (shooting stars)
 3. Comet: Small icy dirt balls that orbit the Sun, burn upon reaching Sun
 4. Stars

- Stars: luminous bodies
- Colour: Depends on temperature
- Group of stars: Constellation



- Largest: Hydra
- Urja Major: Sapta Rishi

- Brightest star in Orion Constellation: Rigel
- Brightest star in night sky (overall): Sirius (Dog Star)
- Closest star to Earth: Sun → Distance from Earth: 150 million km (1.5×10^8 km)



After Sun, it is Proxima Centuri

- Light Year/Parsec: celestial distances
- 1 LY: 9.46×10^{12} km
- 1 Parsec: 3.26 LY

Sun

- India's first Solar Mission



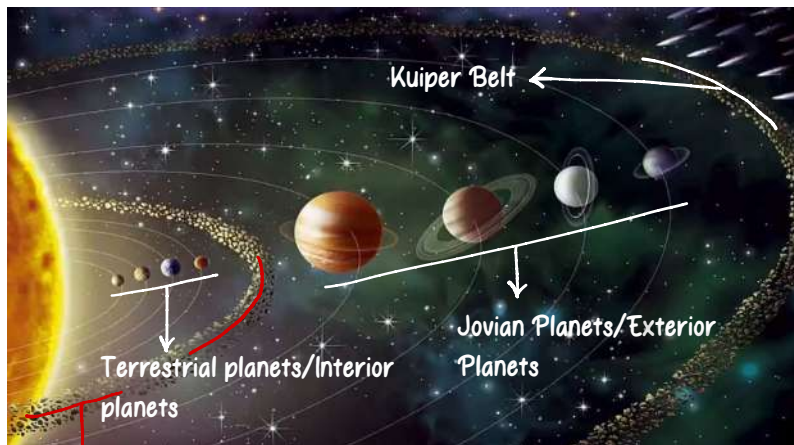
- ADITYA L1 mission ISRO, India

- The only star in our solar system and powerhouse of solar system
- Composed of Hydrogen (73%), Helium (25%) and other metals
- Carries 99% mass of our solar system
- Approx 109 times of Earth
- Takes 8 minutes 30 seconds for light at speed of 3 lakh km/sec to reach Earth
- Temperature at surface = 5800 K/5600 C
- Temperature at centre = 15.7 million K
- Outer layer: CORONA

Moon

- Earth's natural satellite
- Non-Luminous
- Radii: 1.74×10^6 km
- Time of Moon's light, takes to reach Earth: 1.26 secs
- Distance b/w Earth and Moon: 3,84,000 km
- Gravity = $\frac{\text{Earth's gravity}}{6}$
- Rotation = Revolution (same)
 - ↓
 - 27.3 days → Only one side of the Moon is visible (far side)
- Rotation: object's spinning motion about its axis
- Revolution: object's orbital motion around another object
- All planets rotate from **West to East** (anti-clockwise) **except Venus and Uranus** (clockwise)





Asteroid Belt: b/w Mars and Jupiter

- Pandit Jasraj becomes the first Indian musician to have a minor planet named after him: Panditjasraj (300128) → Derived from his date of birth, 28 Jan 1930

• Characteristics of Terrestrial Planets



Revolves around planets



Jupiter

Saturn



Uranus

Neptune

1st Planet: Mercury

- Closest planet to Sun
- Smallest planet in solar system
- Diameter: 4900 km
- Fastest planet, takes 88 days to complete revolution around Sun
- Planet with no satellite
- Planet with no water and gases like Nitrogen, Hydrogen, Oxygen, and Carbon Dioxide

2nd Planet: Venus

- Hottest planet in solar system: traps the gas easily, has thick clouds of H_2SO_4 and CO_2
- Brightest planet in Solar System, also known as "Evening Star" and "Morning Star"
- No satellite/Moon
- Also known as "Earth's Twin" due to similar mass and size
- Rotates clockwise

3rd Planet: Earth

- the only planet to give support to life
- Also known as "Blue Planet": 70% water
- It has one satellite: Moon
- Densest in the entire solar system

4th Planet: Mars

- Known as "Red Planet": rich in iron oxide (red soil)
- Second smallest planet in solar system
- Two natural moons: Phobos and Deimos
- Largest Volcano and tallest mountain of Mars: Olympus Mons

5th Planet: Jupiter

- Largest planet with shortest rotation- 10 hours
- Atmosphere filled with: Hydrogen, Helium, other gases
- Third brightest after Moon and Venus
- At present total moons: 95 moons at present
- Largest satellites: Io, Europa, Ganymede (largest among all), Callisto (all discovered by Galileo)
- Has unclear ring around it

6th Planet: Saturn

- Second largest planet
- Has bright and concentric rings made of tiny rocks, gas, dust, ice
- It is the least dense planet
- Has 146 moons at present (the maximum)
- Largest satellite: Titan
- 1655: Huygenes (discover Saturn's rings)
- 1675: Cassini (discovered gap b/w rings)

→ Cassini divisions

7th Planet: Uranus

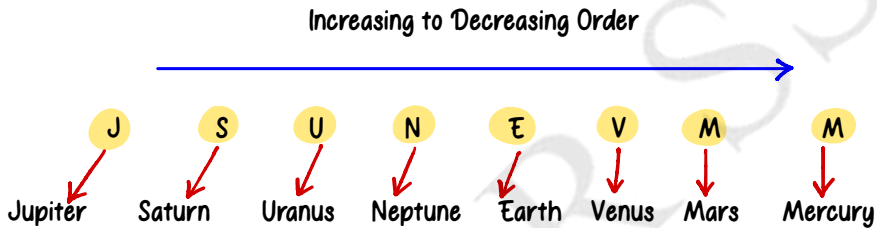
- It is greenish in colour: "Green Planet" due to presence of Methane (CH_4)
- Discovered by William Herschel in 1781
- Known as "Ice Giant"
- Atmosphere has: Hydrogen, Helium, Water, Ammonia, Methane
- Rotates clockwise like Venus
- Coldest planet
- Its is tilted to 98° at its axis- Rolling/Lopsided Planet

8th Planet: Neptune

- Farthest planet
- It is also "Ice Giant"
- Atmosphere composed of: Hydrogen, Helium
- Bluish in colour due to Methane
- Fourth largest planet and third most massive planet
- Discovered by: Johann Galle and Urbain Le Verrier in 1846 (Only planet found by Mathematical Predictions)
- Has 14 satellites, famous moon: Triton

Pluto

- No more a planet in 2006 by International Astronomical Union (IAU)
- It is known as dwarf planet and is a member of Kuiper Belt
- Kuiper Belt is a spherical boundary outside the orbit of Neptune containing a number of asteroids, rocks and comets





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Geography

Latitude - Longitude Rotation - Revolution

Lecture :- 2

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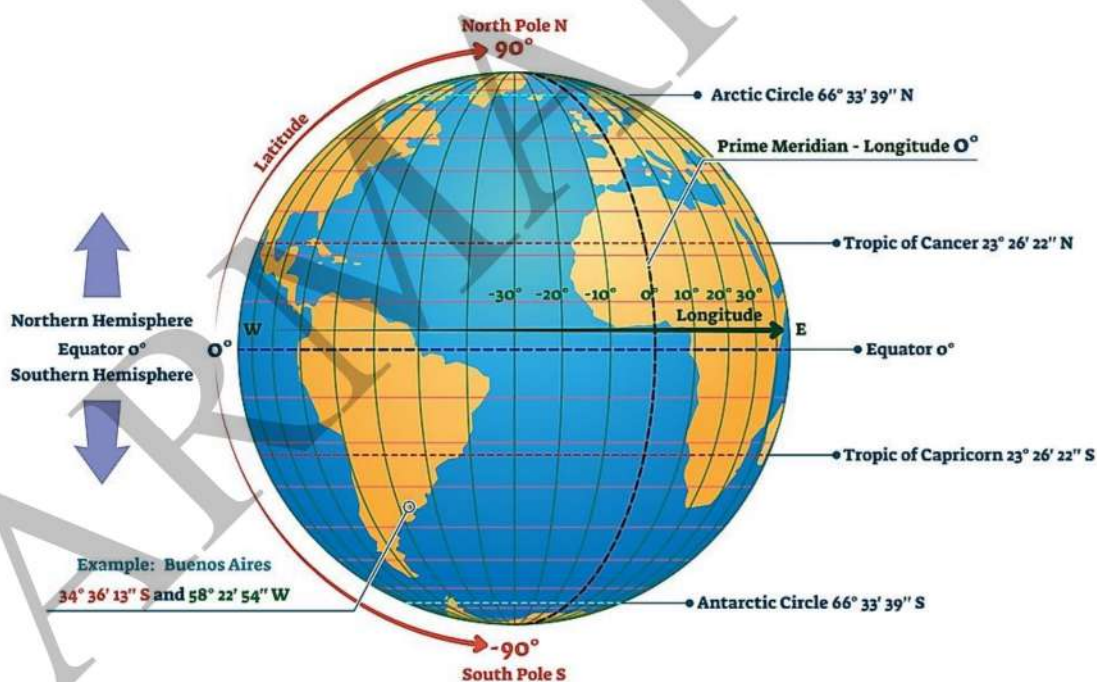


Click on the icon.

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LONGITUDE AND LATITUDE ROTATION AND REVOLUTION



Fundamentals of Earth



Phase :-1

- Age of Earth (पृथ्वी की आयु)
- Shape of Earth (पृथ्वी का आकार)



Phase:-2

- Axis and Orbit (अक्ष और कक्षा)
- Latitudes and Longitudes (अक्षांस और देशांतर)



Phase:-3

- Concept of Time (समय की अवधारणा)
- Seasons on Earth (पृथ्वी पर ऋतुएं)

Phase 4: Eclipse

Age of Earth

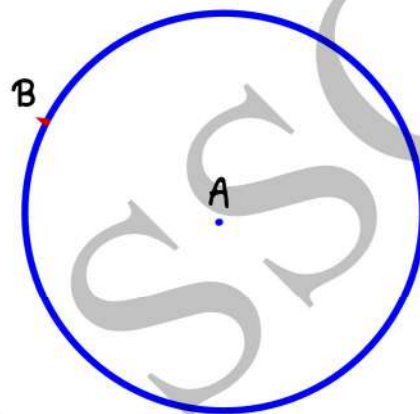
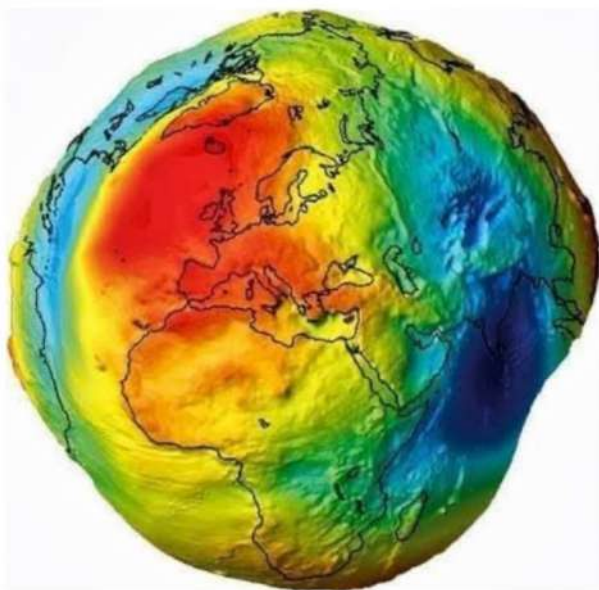
Technique used: Radioactive dating $\rightarrow$ invented by Ernst Rutherford (1905)

Types of Dating

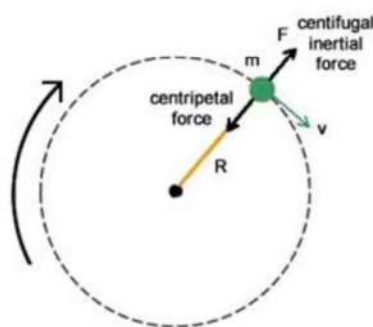
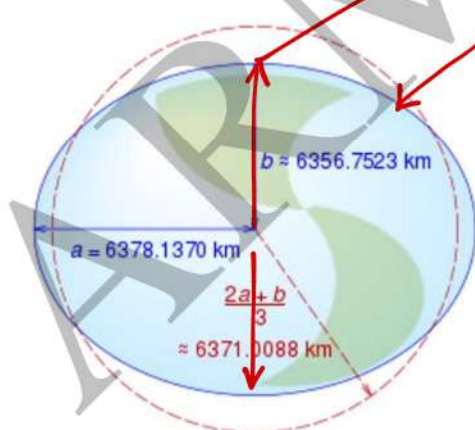
1. Uranium-lead dating method (oldest rocks)
2. Potassium-argon method
3. Rubidium-strontium method
4. Radiocarbon dating method
5. Chlorine-36 dating method
6. Carbon-dating (C^{14}) (latest rocks)



Shape of Earth



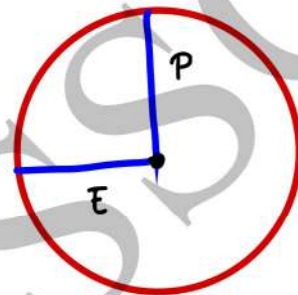
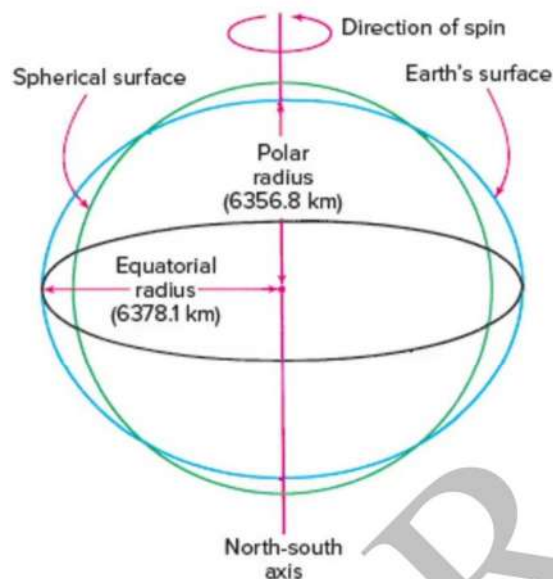
- Shape of Earth is Geoid or Oblate Spheroid (a little flat from top and bottom)
- Reason: more Centrifugal Force at Equator bulges earth at Centre and Gravitation force at poles pushes surface towards centre due gravitational force towards the centre, it flat in top and bottom



When a body revolves, two types of forces is applicable

- Centripetal Force: towards the axis of rotation or centre of curvature (inside)
- Centrifugal Force: directed away from the centre of the circle

Radius of Earth



- Equatorial Radius: 6378 km
- Polar Radius: 6357 km
- Mean Radius: 6371

Why polar radius < Equatorial radius?

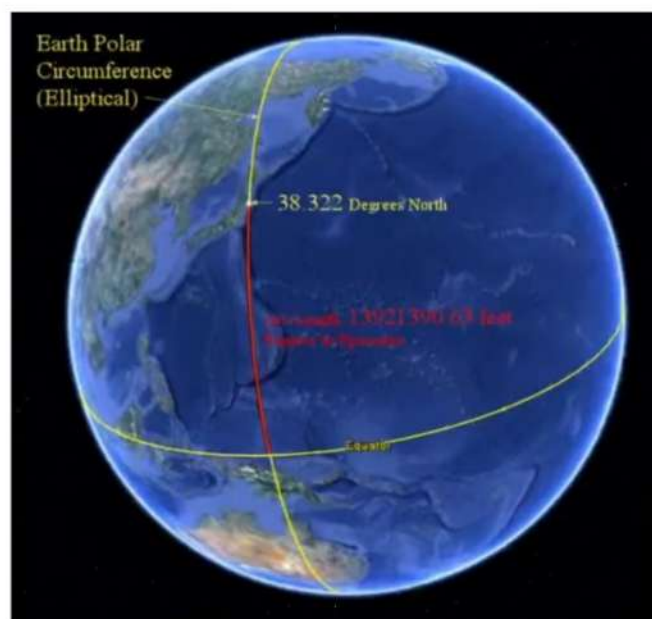
- Ans: Earth is bulged at the equator and flattened at the poles

Circumference of the Earth

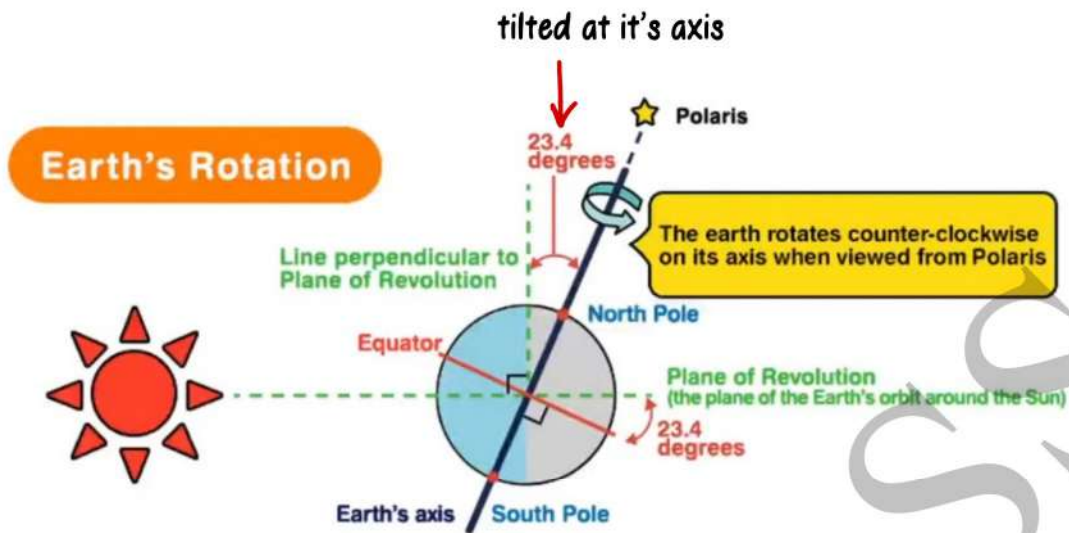
- Polar: 40,007 km
- Equatorial: 40,075 km
- Mean: 40,040 km

Why poles circumference < Equatorial?

- Earth is bulged at equator and flattened at the poles

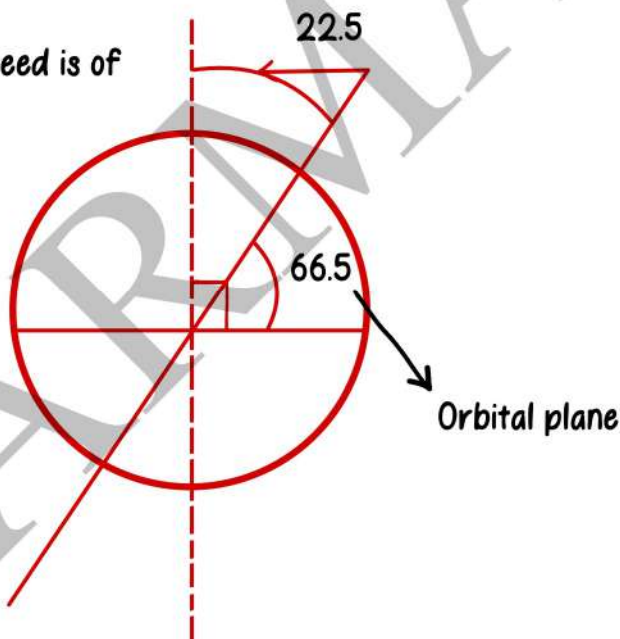


Rotation of Earth

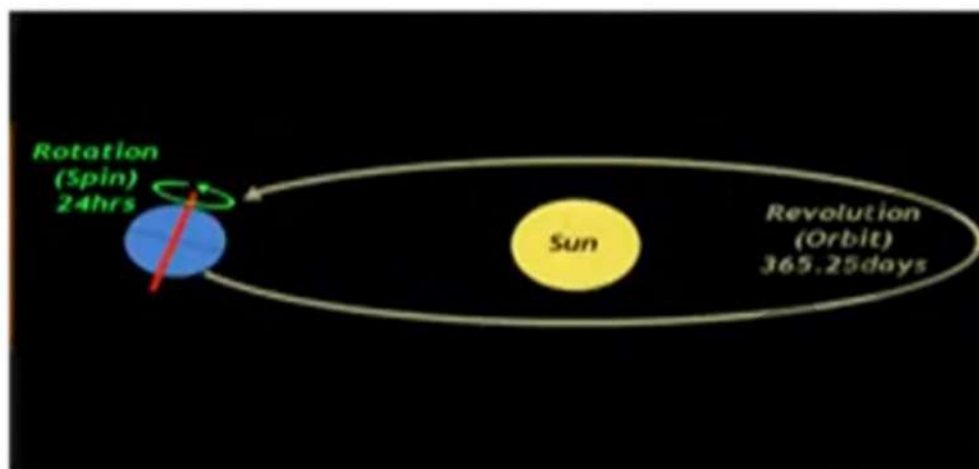


- Rotation: spinning on its own axis
- One rotation of Earth: 23 hour 56 mins 4 sec
- Direction: West to East
- Rotational Speed is maximum at Equator and minimum at Poles

Fastest rotation speed is of Jupiter

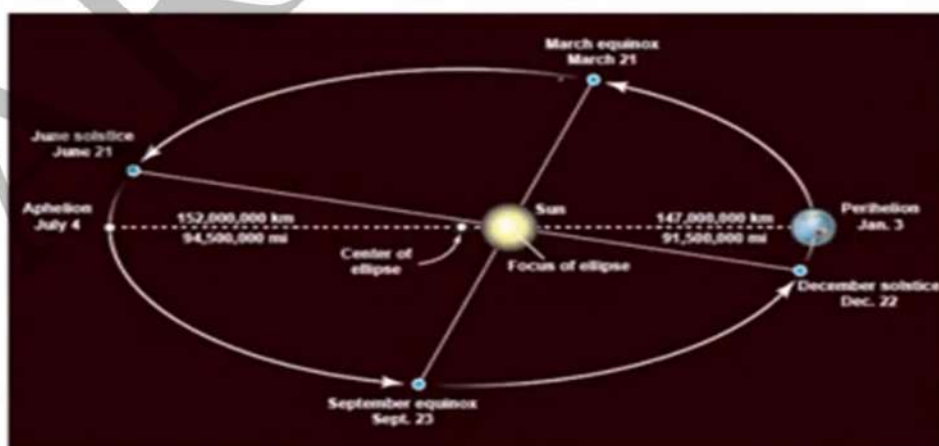


Revolution of Earth



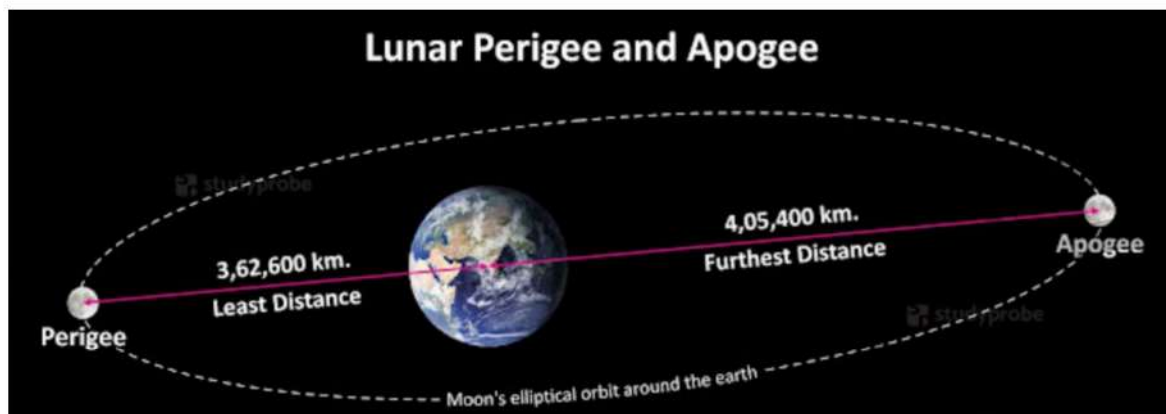
- Revolving around the Sun in Elliptical orbit
 - One revolution: 365 days 6 hours 9 minutes and 9 sec
 - Orbital speed: 29.8 km/sec
 - Max orbital speed: Mercury
 - Min orbital speed: Neptune
- $6 \times 4 = 24 \text{ hrs} \rightarrow \text{Leap year concept (366 days)}$

Distance from the Sun

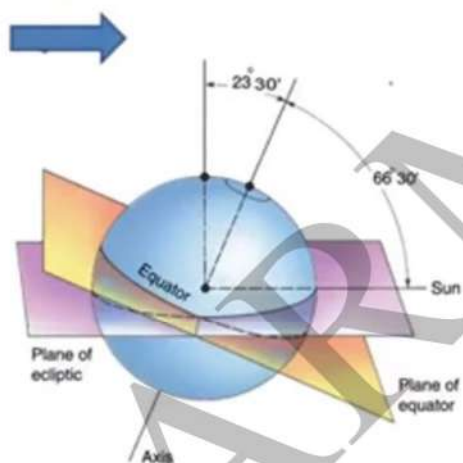


- When nearest to Sun: Perihelion (January 3rd – 14,75,00,000 km)
- When farthest from Sun: Aphelion (July 4 – 15,25,00,000 km)

- **Perigee:** the point of moon's orbit when it is closest to Earth
- **Apogee:** When moon is farthest from Earth



Inclination of the Earth's axis



- **Axial Inclination:** Inclination of Earth on its axis = $23 \frac{1}{2}^{\circ}$
- **Orbital Inclination:** Inclination of Earth on its orbital plane = $66 \frac{1}{2}^{\circ}$

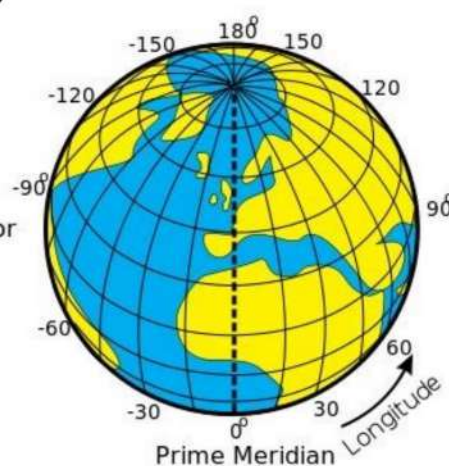
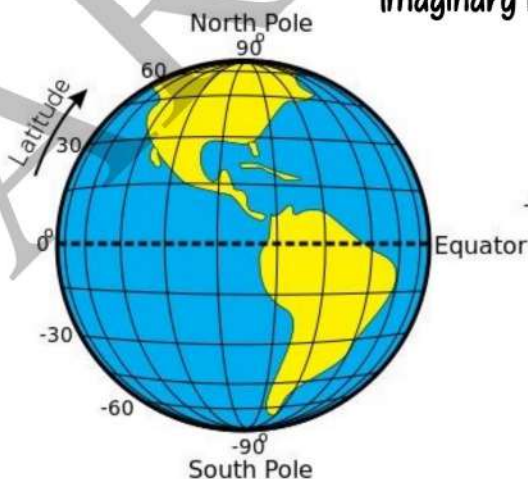
Hemisphere



- Equal division of Earth in two parts
- Equator: divides the globe horizontally into 2 equal parts – Northern and Southern Hemisphere
- Prime Meridian and International Date Line: divides the globe vertically – Eastern and Western Hemisphere

Latitude and Longitude

Imaginary lines

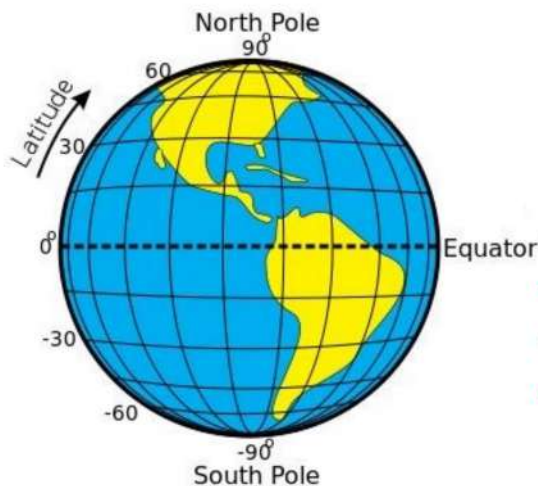


Horizontal lines



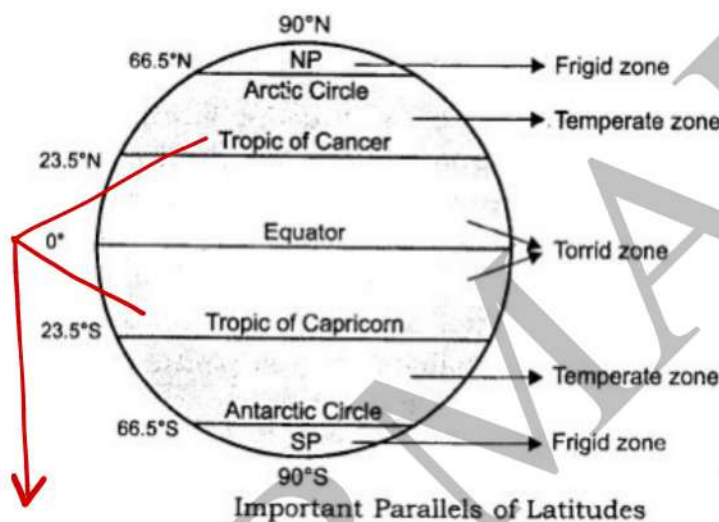
Vertical lines





Latitude

- Imaginary horizontal lines on the globe that run from East to West
- Angular Distance of a place from the equator
- 1 degree of latitude = 111 km (approx)
- Total latitudes: 181
- Distance b/w each latitude is same

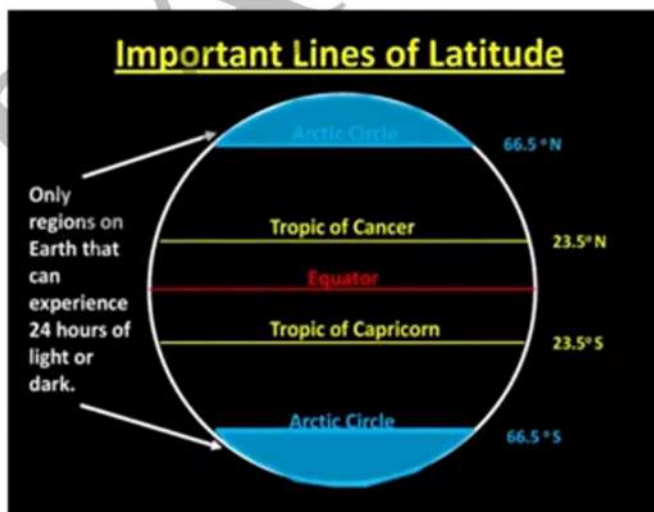


Sunlight does not fall beyond these tropics

Important Latitudes:

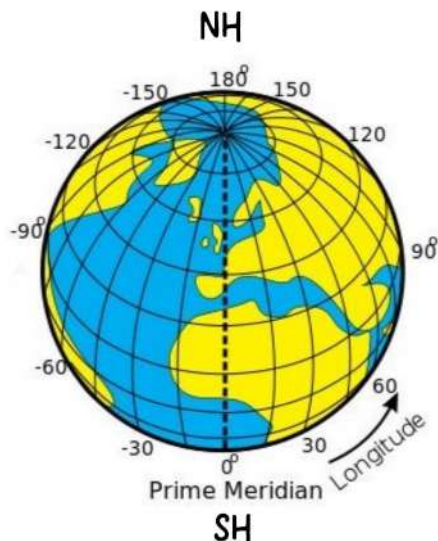
- 0 : Equator
- $23\frac{1}{2}^{\circ}$ N: Tropic of Cancer
- $66\frac{1}{2}^{\circ}$ N: Arctic Circle
- $23\frac{1}{2}^{\circ}$ S: Tropic of Capricorn
- $66\frac{1}{2}^{\circ}$ S: Antarctic Circle
- Largest latitude: Equator
- Smallest latitude: Poles (North and South)

Important Lines of Latitude



Uses

1. In Climatology:
 - Temperature zones, wind
 - Responsible for Pressure System
 - Planetary Winds System
2. Location of place



Longitudes

- Imaginary vertical lines over the globe that run North to South
- Angular Distance of a plane from Prime Meridian
- Distance from each longitude varies from poles towards equator
- Least distance at poles and maximum distance at equator: 111.32 km
- Total longitudes: 360

*

- All longitudes divide Earth into 2 equal parts
- All longitudes are Great Circle (circle in case of longitudes)



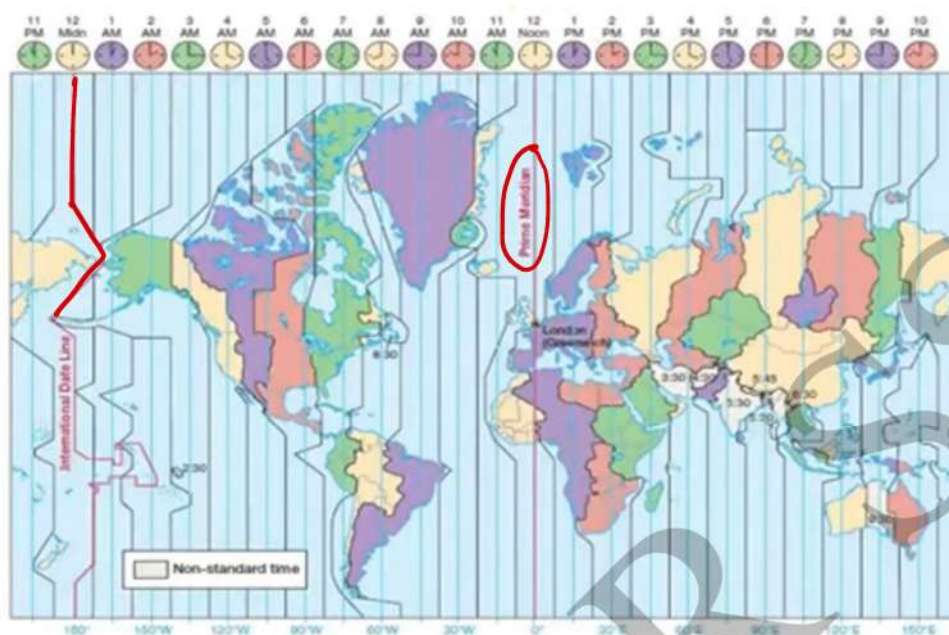
Important Meridians

- Prime Meridian: 0 degree longitude (passes from Greenwich, London)
- International Date Line: 180 degree Meridian



Zig-Zag lines

International Date Line



Prime Meridian



- It passes through Greenwich in London
- Countries: 8
 - UK
 - France
 - Spain
 - Algeria
 - Mali
 - Burkina Faso
 - Togo
 - Ghana
- TRICK: BSF GAME in TOGO Kingdom



$$360^\circ = 24 \text{ hrs}$$

$$\frac{360^\circ}{24} = 15^\circ = 1 \text{ hr}$$

$$15^\circ = 1 \text{ hr}$$

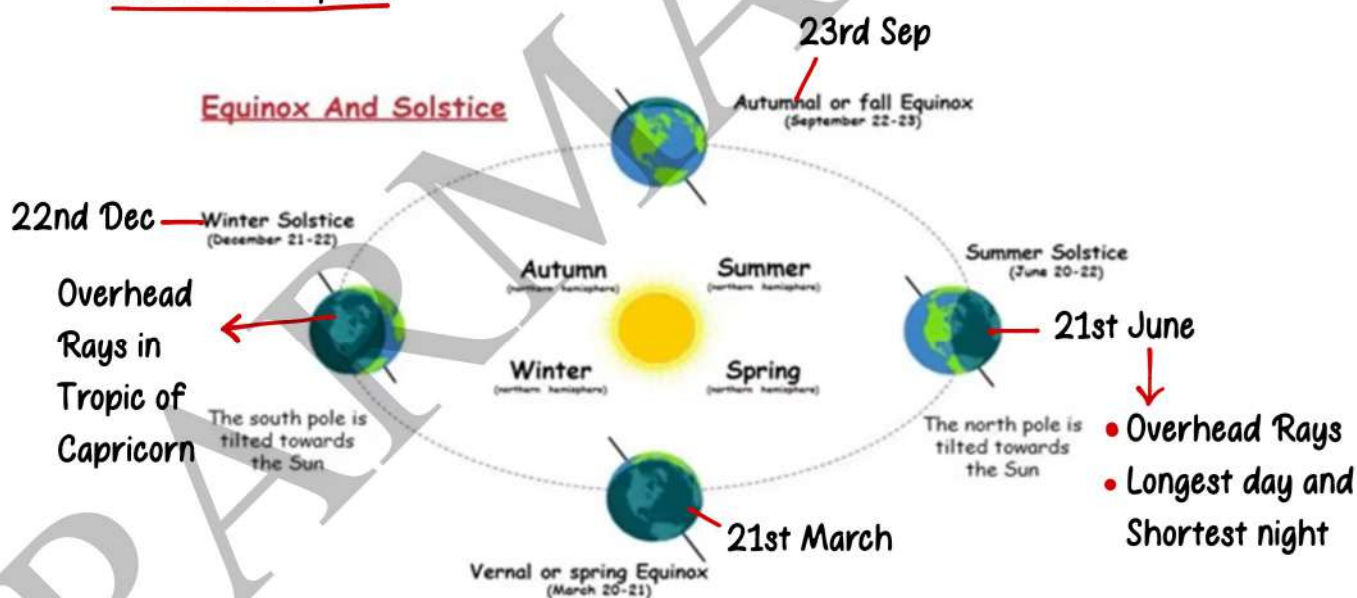
$$15^\circ = 60 \text{ mins}$$

$$1^\circ = \frac{60}{15} = 4 \text{ mins}$$

- Prime Meridian will increase the time by 1 hr
- Moving each 15° towards West of Prime
- Meridian will decrease the time by 1 hr

Solstice and Equinox

Equinox And Solstice



- Day and Night: due to Rotation

- Seasons:

1. Revolution
2. Tilt

Solstice

Summer - June 21

1. Vertical rays on Tropic of Cancer
2. Northern Hemisphere gets more heat
3. Continuous sun rays on North Pole for 6 months, continuous days
4. known as **Kark Sankranthi**

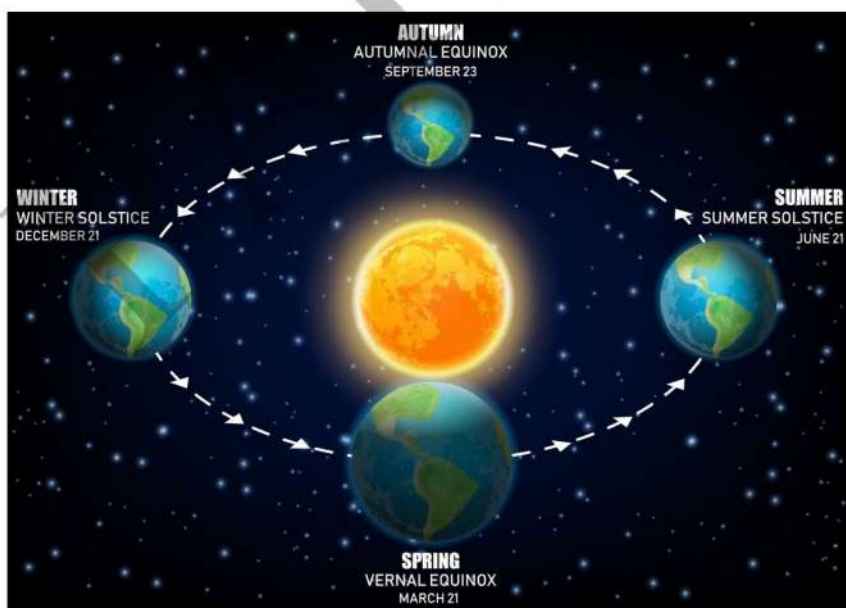
Insolation: incoming solar radiations

Winter - Dec 22

1. Vertical rays on Tropic of Capricorn
2. Southern Hemisphere gets more heat
3. Continuous Sun rays on South Pole for 6 months, continuous daylight
4. known as **Makar Sankranthi**

Equinox

- Direct rays of the Sun fall on the Equator
- At this position neither of the poles is tilted towards the Sun
- So, the entire Earth experiences Equal days and nights



Vernal Equinox

- **March 21**: It is spring in the NH and autumn in the SH

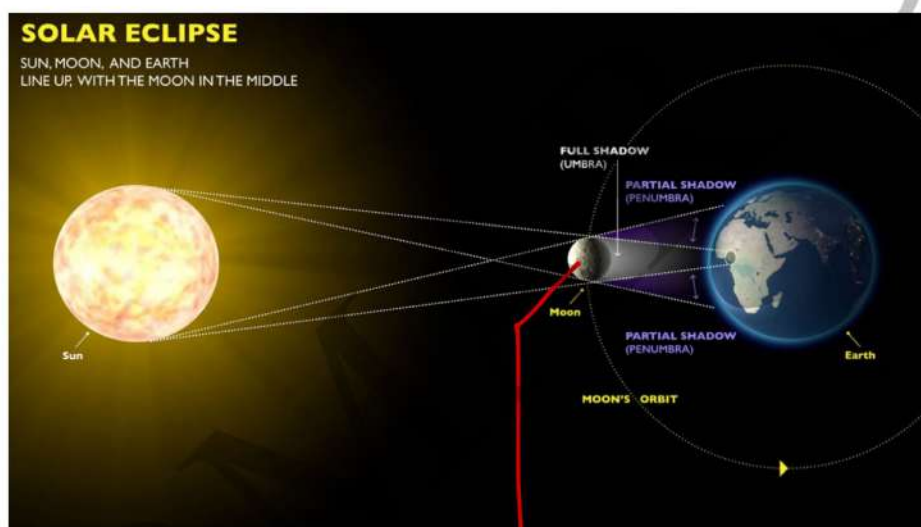
Autumnal Equinox

- **Sep 23**: it is autumn in NH and spring in SH



- Sun (at its constant position) is obscured by the moon

↓
New Moon
- Amavasya



↓
Moon is blocking Sun's light

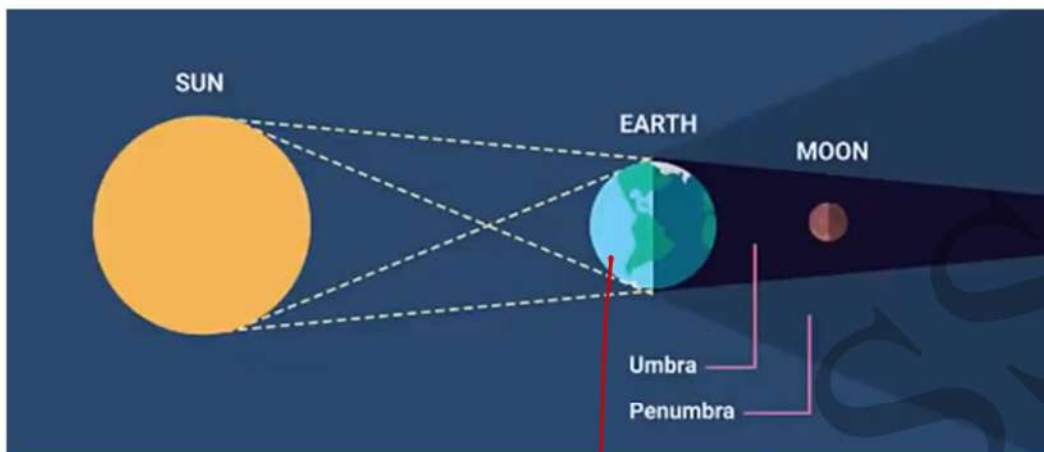


Total Solar Eclipse

Annular Solar Eclipse

Partial Solar Eclipse

Lunar Eclipse

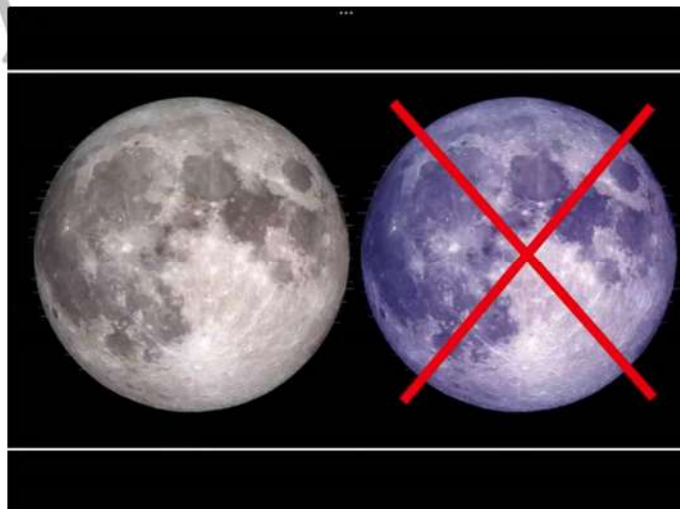


- Full Moon condition- Purnima

Earth blocks Sun's light
(light refraction) → scatters more
causing blue colour
light to vanish and red → scatters less
light to reach moon

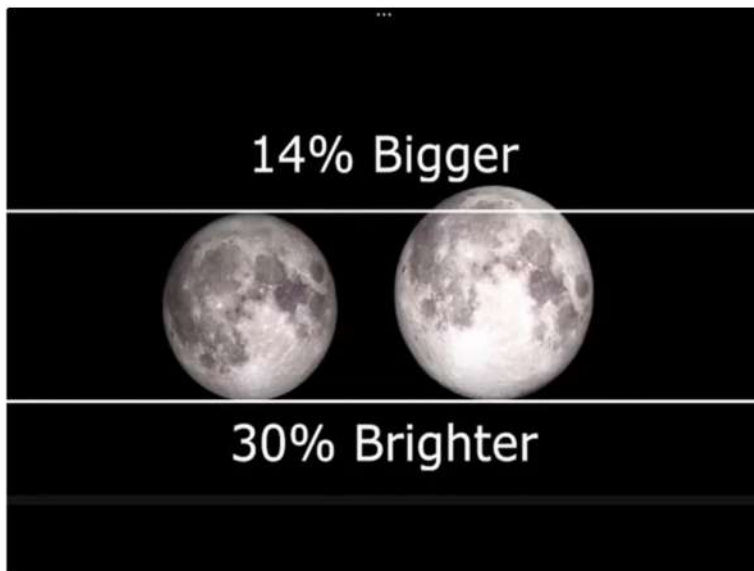


Red Moon



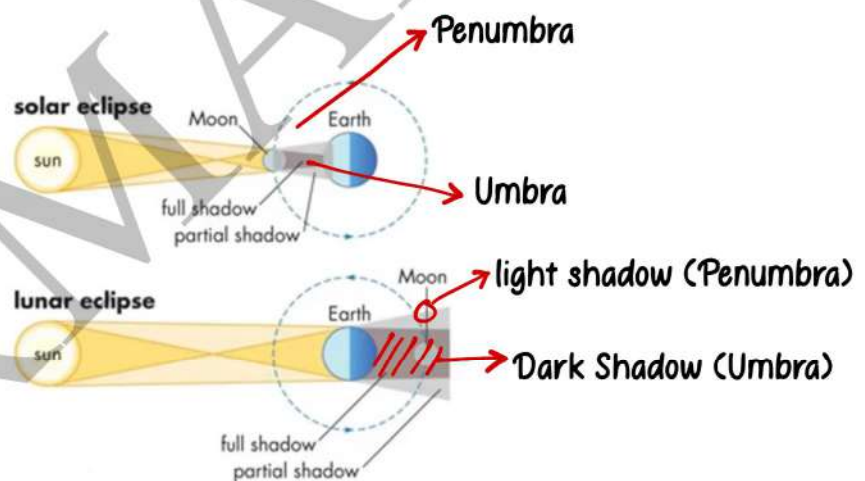
Blue moon

↓
2 full moon in a month



- Lunar Eclipse + Perigee → Moon appears bigger than its normal size

↓
Super Moon condition





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PARMAR'S GK BATCH

Geography

Earth's Interior and Plate Tectonics

Lecture :- 3



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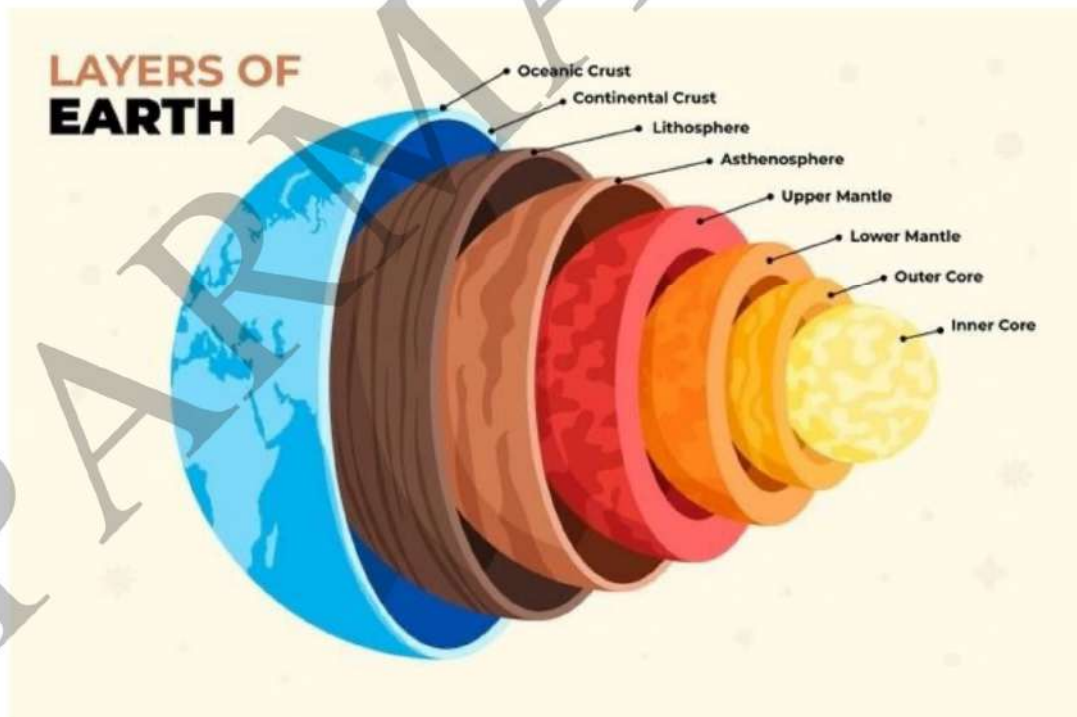


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Earth's Interior & Plate Tectonic





Phase -1

- Origin of Earth(पृथ्वी की उत्पत्ति)
- Plate tectonic theory (प्लेट टेक्टोनिक सिद्धांत)



Phase -2

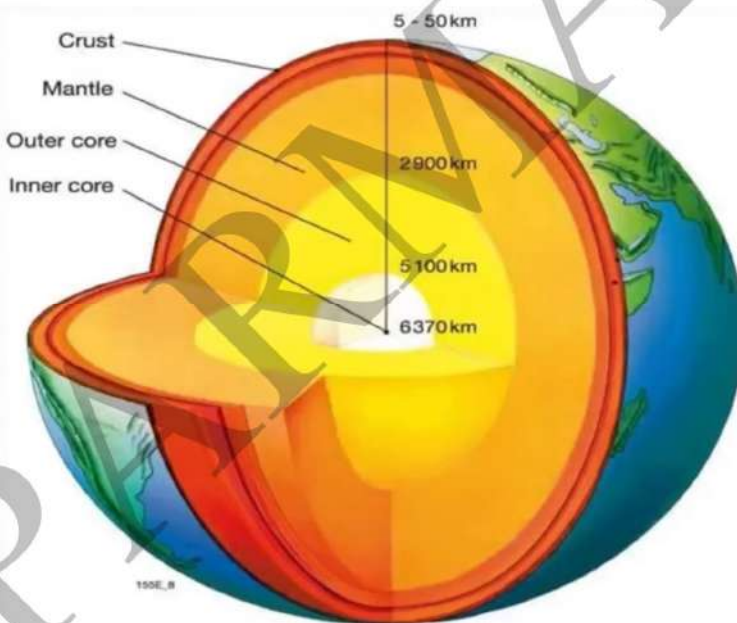
- Interior of Earth(पृथ्वी का आंतरिक भाग)
- Earthquake & Volcanoes(भूकंप और ज्वालामुखी)



Phase -3

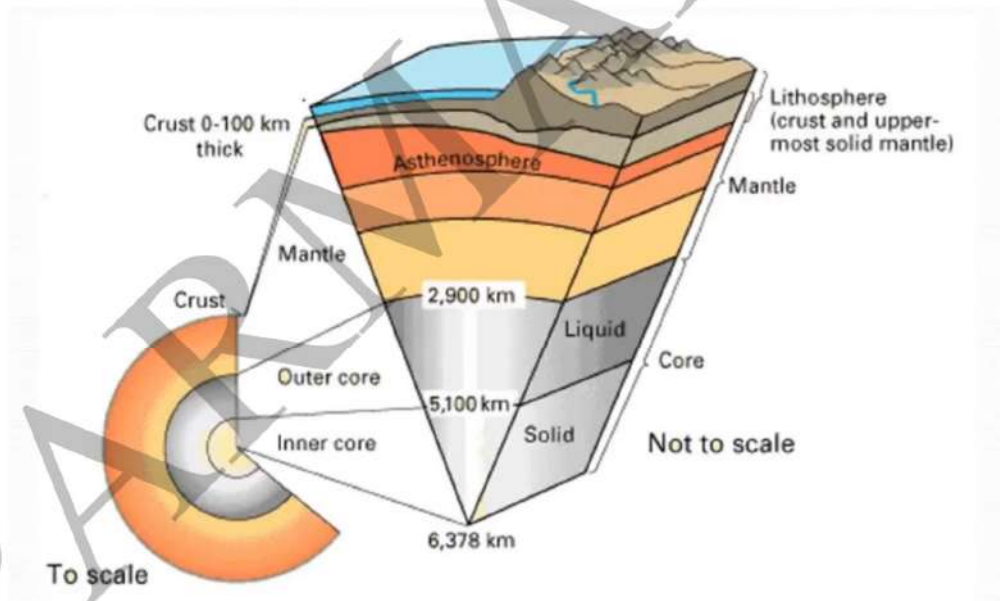
- PLATE MOVEMENTS

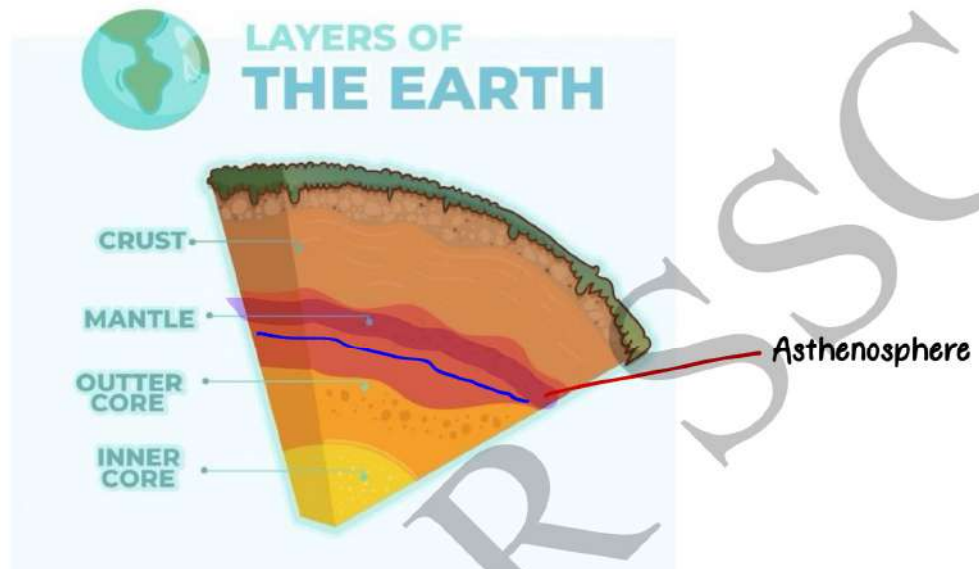
Earth's Interior



Four method's to know Earth's interior:

1. Temperature - indirect source
2. Volcanoes and rock - direct source
3. Meteorites - indirect source
4. Earthquakes - indirect source





Crust: made of Silica and Aluminium layer (SiAl)

- Thickness: 5-70 km

Two divisions:

1. Continental Crust:

- land part of crust
- 30 km (thick/lighter)
- made of Granitic rock

2. Oceanic Crust:

- water part of crust
- 5 km (thick/denser)
- made of Basaltic rock

Composition of Earth's crust:

- O → 46.4%
- Si → 28%
- Al → 8% (most abundant metal in crust)
- Fe → 5% (2nd most abundant)

Mantle: made of Silica and Magnesium (SiMa)

- Thickness: 2900 km
- Top layer: Solid form

Two divisions:

1. Upper Mantle
2. Lower Mantle

- Asthenosphere: semi-molten form (plastic form)

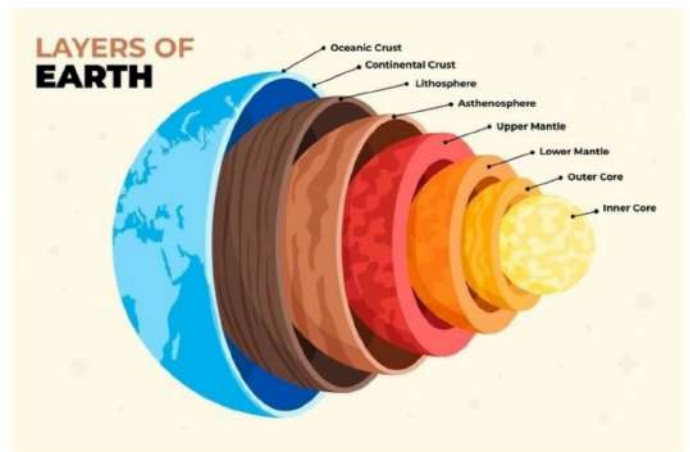
Core: made of Nickel and Iron (NiFe)

Two divisions:

1. Inner Core: Solid form - 2200 km
2. Outer Core: liquid form (shows magnetic properties) - 1300 km

	<u>Crust</u>	<u>Mantle</u>	<u>Core</u>
<u>By Volume</u>	1%	84%	15%
<u>By Mass</u>	1%	68%	31%

- Lithosphere: Crust + Upper solid part of Mantle - thickness: 10-200 km
- Asthenosphere is not part of Lithosphere



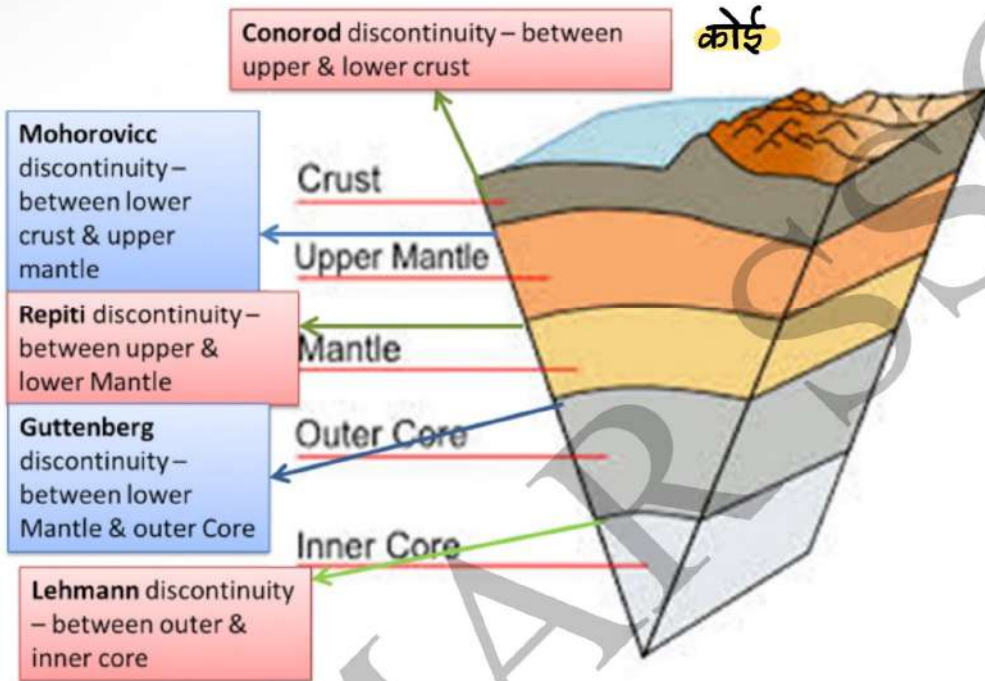
Earth's Discontinuity

मुझे

Red

मुलाख

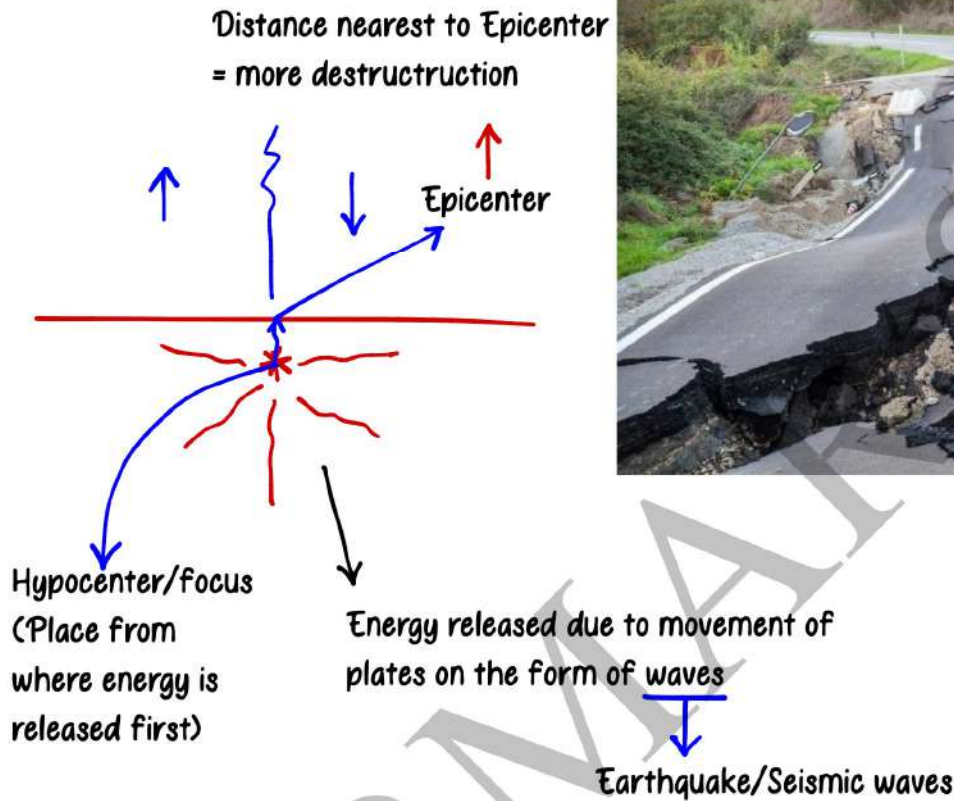
लादो



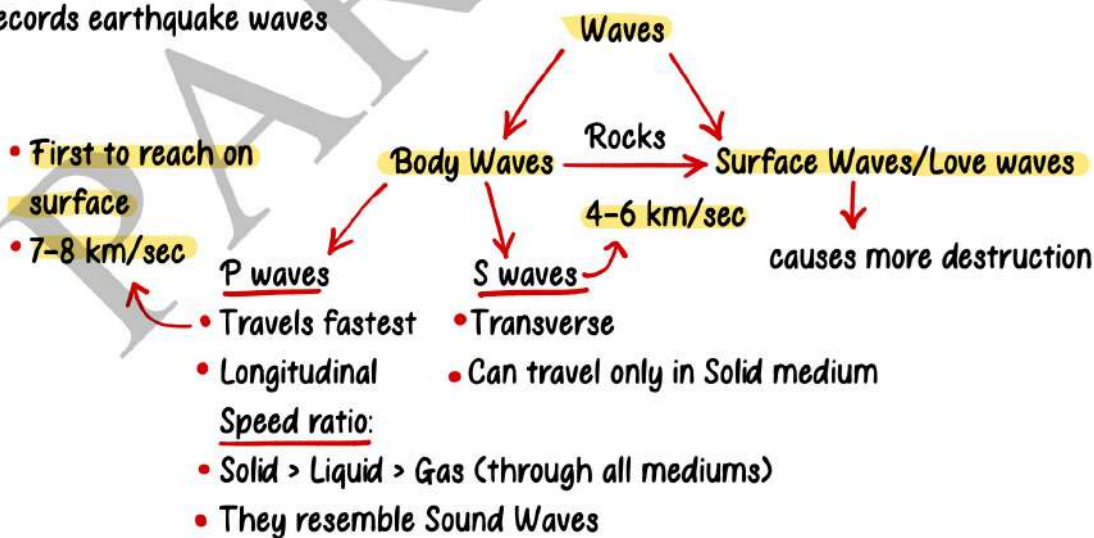
S. No	Discontinuity	Layers	Depth
1.	Conrad	Outer and Inner Crust	45 km
2.	Moho	Crust and Mantle	100 km
3.	Repiti	Inner Crust and Outer Mantle	
4.	Gutenberg-Weichart	Inner Crust and Asthenosphere	700 km
		Outer Mantle and Inner Mantle	
		Mantle and Core	2900 km
		Inner Mantle and Outer Core	
5.	Lehmann	Outer Core and Inner Core	5200 km

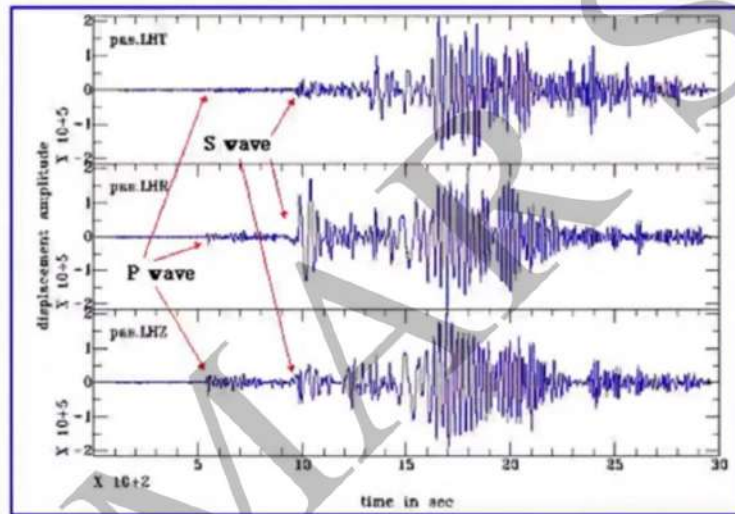
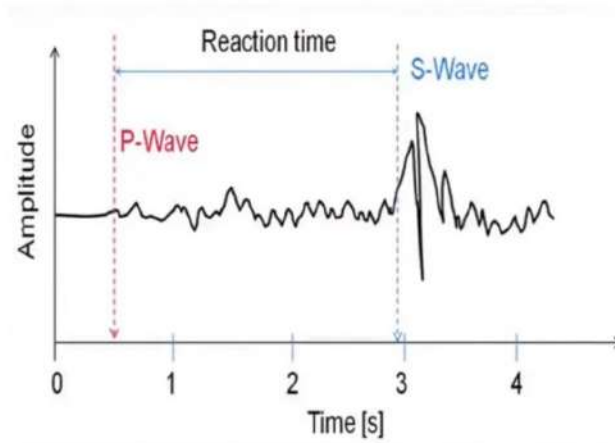
Earthquake

- An Earthquake is intense shaking of Earth's surface, which causes shifting of Earth's plate



- **Seismograph**: an instrument that records earthquake waves





S waves



- creates Crest and Trough

P waves

- creates Compression and Rarefaction
- causes stretching and squeezing



Scales to measure Earthquake

EARTHQUAKE MAGNITUDE SCALE



Richter Scale

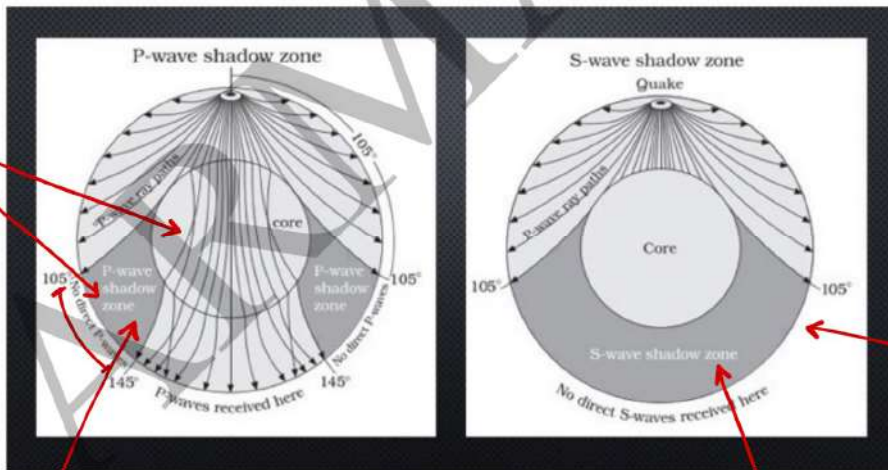
- Instrument to measure **magnitude of Earthquake**
- Magnitude: **0-10**
- It is a **limitless scale**

Mercalli Scale

- Instrument to measure **intensity of Earthquake**
- Magnitude: **1-11**

Shadow zone of waves

speed slow



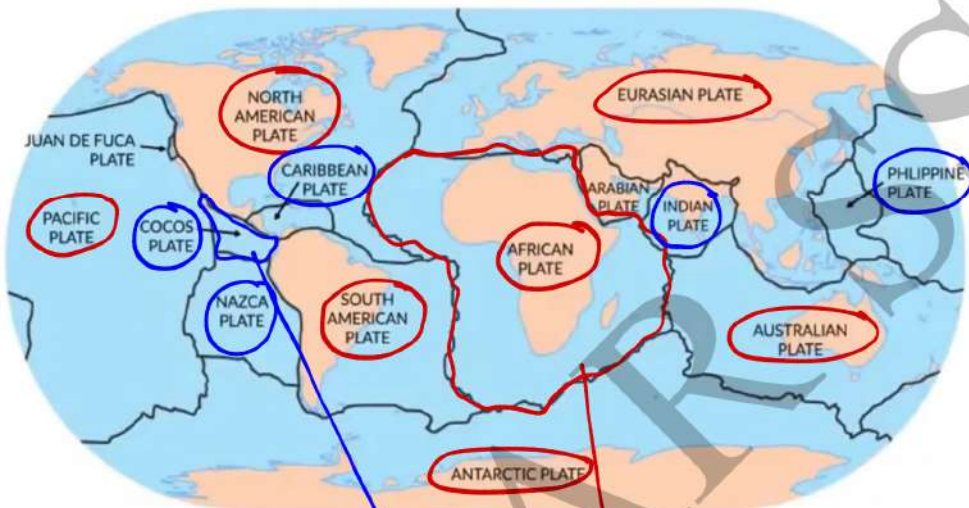
Liquid medium

Liquid medium

- Large shadow zone
- **40% of Earth's surface (not recorded)**

Tectonic plates

- Lithosphere makes plates comprising Crust and upper solid part of Mantle
- 7 Major + few minor plates



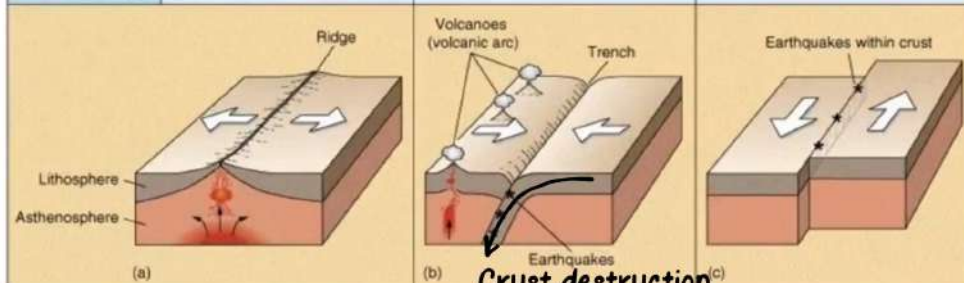
Large boundary: major plate

Minor boundary: minor plate

- Major plates marked in red
- Minor plates marked in blue

Different types of plate boundaries

Type of Margin	Divergent	Convergent	Transform
Motion	Spreading	Subduction	Lateral sliding
Effect	Constructive (oceanic lithosphere created)	Destructive (oceanic lithosphere destroyed)	Conservative (lithosphere neither created or destroyed)
Topography	Ridge/Rift	Trench	No major effect
Volcanic activity?	Yes	Yes	No



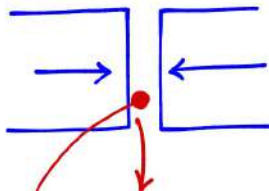
(a) Ridge
Lithosphere
Asthenosphere

(b) Volcanoes (volcanic arc)
Trench
Earthquakes

(c) Earthquakes within crust

Crust destruction

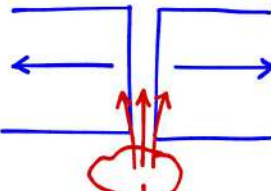
Convergent



Denser one subducts

Subduction zone created
Asthenosphere melts
the crust causing crust
destruction

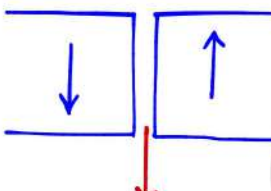
Divergent



distance created, causing lava to
explode

New crust formation -
Constructive boundary

Conservative



Lateral Sliding
Conservative boundary

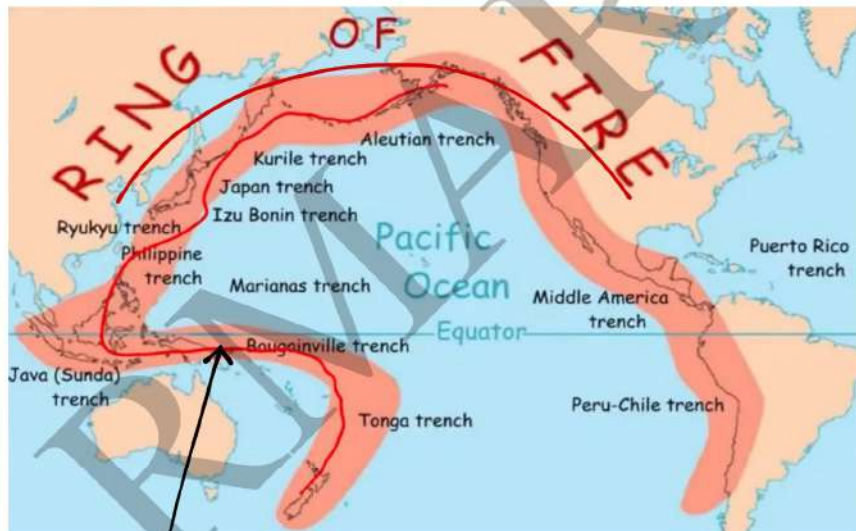


Plate boundary created,
these places are called-
**Ring of Fire in Pacific
Ocean**

Force behind plate movement:

- Heat generated within the Earth due to Radioactive decay and Residual heat

Seismic Zones in India-

Seismic Zone
Map of India: -2002

About 59 percent of the land area of India is liable to seismic hazard damage

Zone	Intensity
Zone V	Very High Risk Zone Area liable to shaking Intensity IX (and above)
Zone IV	High Risk Zone Intensity VIII
Zone III	Moderate Risk Zone Intensity VII
Zone II	Low Risk Zone VI (and lower)

Seismic zones in India

